LUNG CANCER DETECTION - USING CNN AND XCEPTION TRANSFER LEARNING

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ABSTRACT

Lung cancer is one of the leading causes of cancer-related deaths worldwide. Early diagnosis is essential for improving survival rates. Manual analysis of CT scan images is time-consuming and prone to errors. This project proposes a Deep Learning-based Lung Cancer Detection System using CNN and Xception Transfer Learning.

The model classifies CT scan images into three categories: Benign, Malignant and Normal. Image preprocessing, data augmentation, and transfer learning techniques are used to improve classification accuracy and support intelligent medical diagnosis.

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INTRODUCTION

- Lung cancer is among the deadliest cancers globally.
- Early detection significantly increases treatment success.
- Manual CT scan examination requires expertise and time.
- Deep Learning enables automated cancer detection.
- CNN and Xception models provide accurate image classification.

LITERATURE SURVEY

S.No	**Title**	**Authors**	**Year**	**Outcomes**
1	Deep Learning for Lung Cancer Detection Using CT Images	Setio et al.	2017	CNN models successfully detected lung nodules with high accuracy.
2	Xception: Deep Learning with Depthwise Separable Convolutions	François Cholleto	2017	Introduced Xception architecture with improved feature extraction performance.
3	Transfer Learning in Medical Image Analysis: A Survey	Cheplygina et al.	2019	Demonstrated effectiveness of transfer learning in medical imaging tasks.

PROBLEM STATEMENT

- Manual CT scan analysis is time-consuming.
- Diagnostic accuracy may vary among radiologists.
- Early-stage lung cancer is difficult to identify.
- Large amounts of imaging data increase workload.
- Need for an automated and reliable detection system.

PROPOSED SOLUTION

- Use CNN and Xception Transfer Learning.
- Perform image preprocessing and augmentation.
- Extract deep features from CT scans.
- Classify scans into four categories:
- **Normal**
- **Benign (non-cancerous)**
- **Malignant (cancerous)**
- Provide accurate diagnostic support.

OBJECTIVES

- Develop an automated lung cancer detection system.
- Improve diagnostic accuracy using transfer learning.
- Reduce radiologist workload.
- Classify CT scans into four classes.
- Assist in early disease diagnosis.

FRAMEWORK (METHODOLOGY)

- Dataset Collection – Gather CT scan images.
- Preprocessing – Resize and normalize images.
- Data Augmentation – Rotate, flip, and zoom images.
- Xception Model – Extract important features.
- CNN Classifier – Classify lung cancer types.
- Training & Validation – Train and evaluate model.
- Prediction – Generate final classification result.

MODULES

- Dataset Collection
- Image Preprocessing
- Data Augmentation
- Xception Feature Extraction
- CNN Classification
- Model Training
- Prediction Output

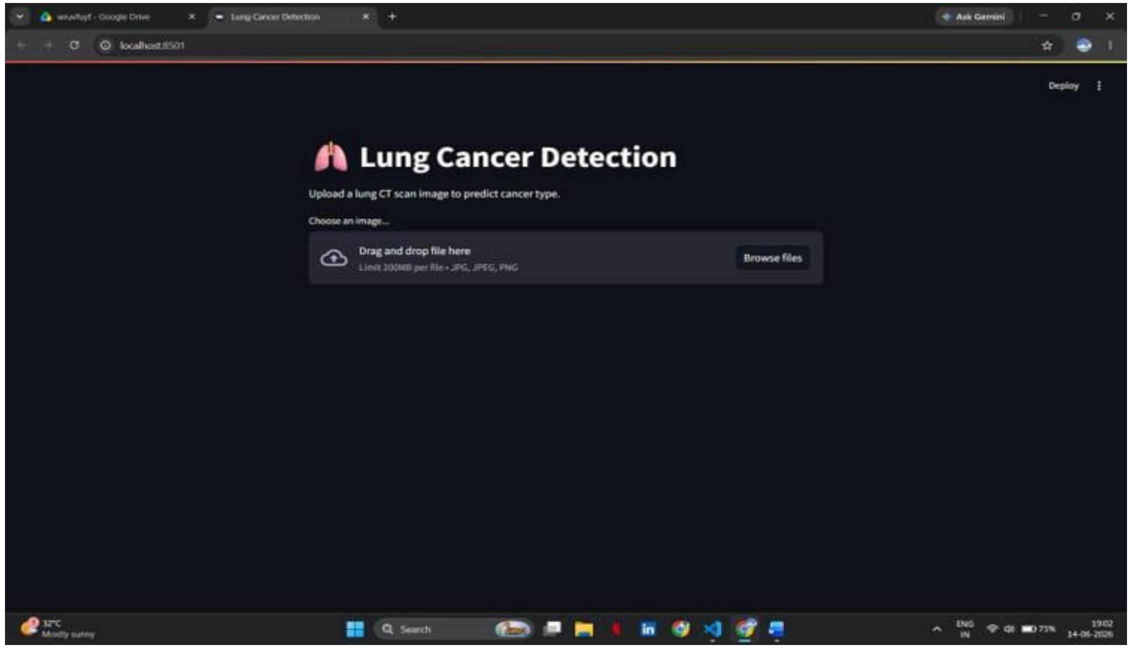
TECH STACK

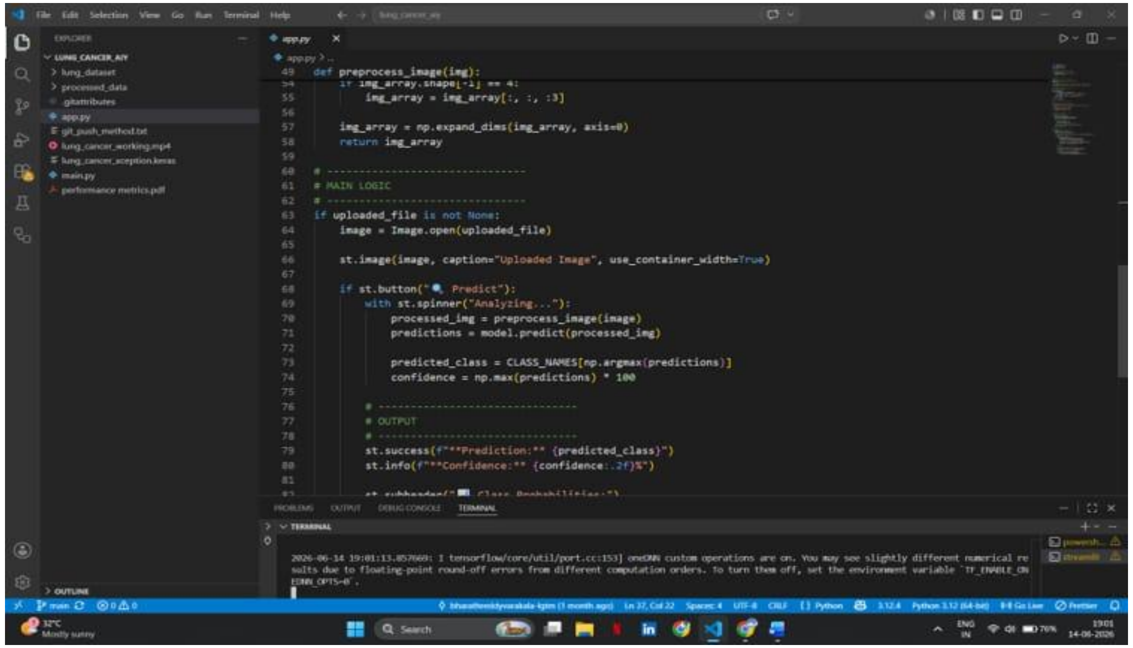
- Language: Python
- Frameworks: TensorFlow, Keras
- Model: CNN + Xception
- Libraries: NumPy, OpenCV, Matplotlib, Scikit-learn
- Tools: Google Colab, Jupyter Notebook, VS Code

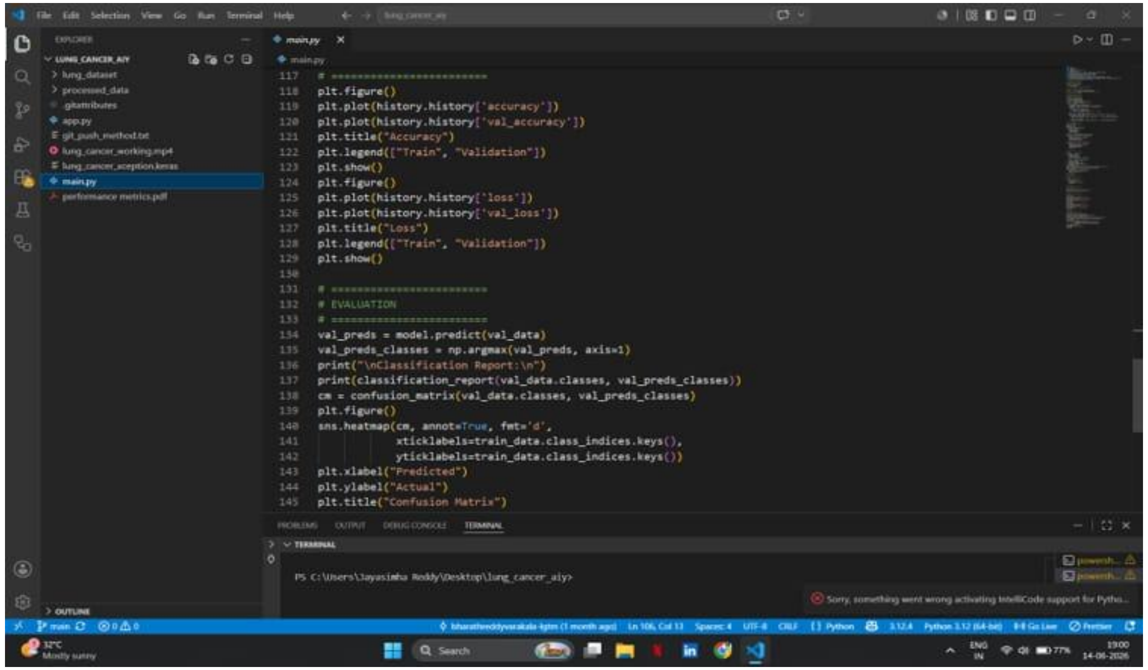
RESULTS

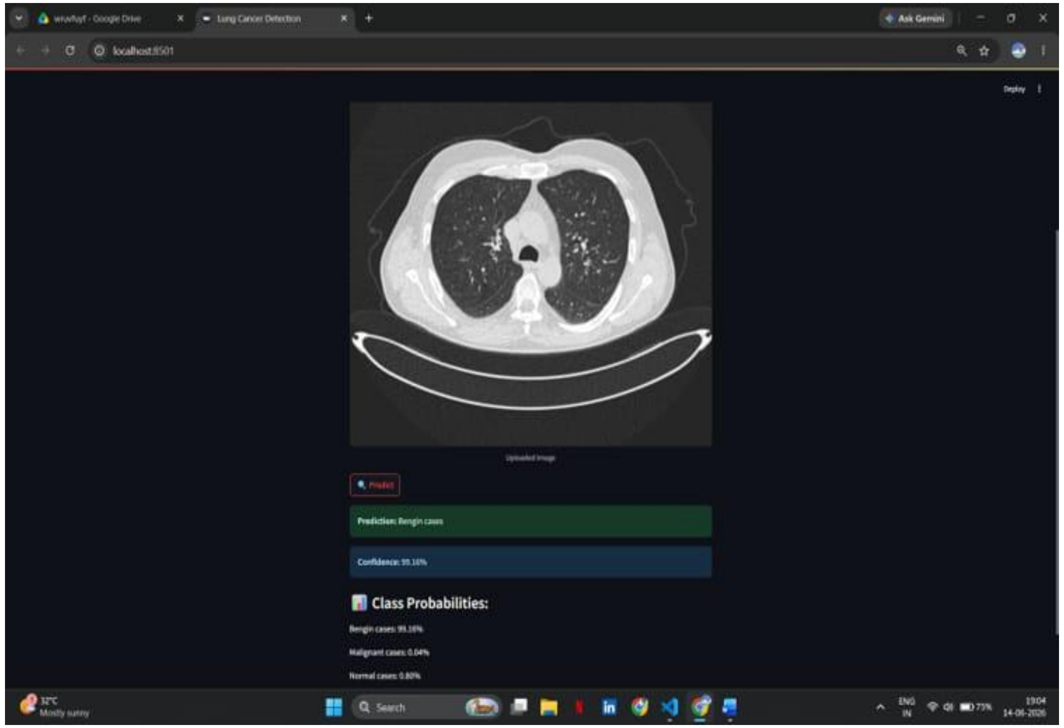
- Successful classification of lung CT scans.
- Improved accuracy using transfer learning.
- Better feature extraction through Xception.
- Reduced chances of misclassification.
- Supports intelligent medical diagnosis.
- **Includes**

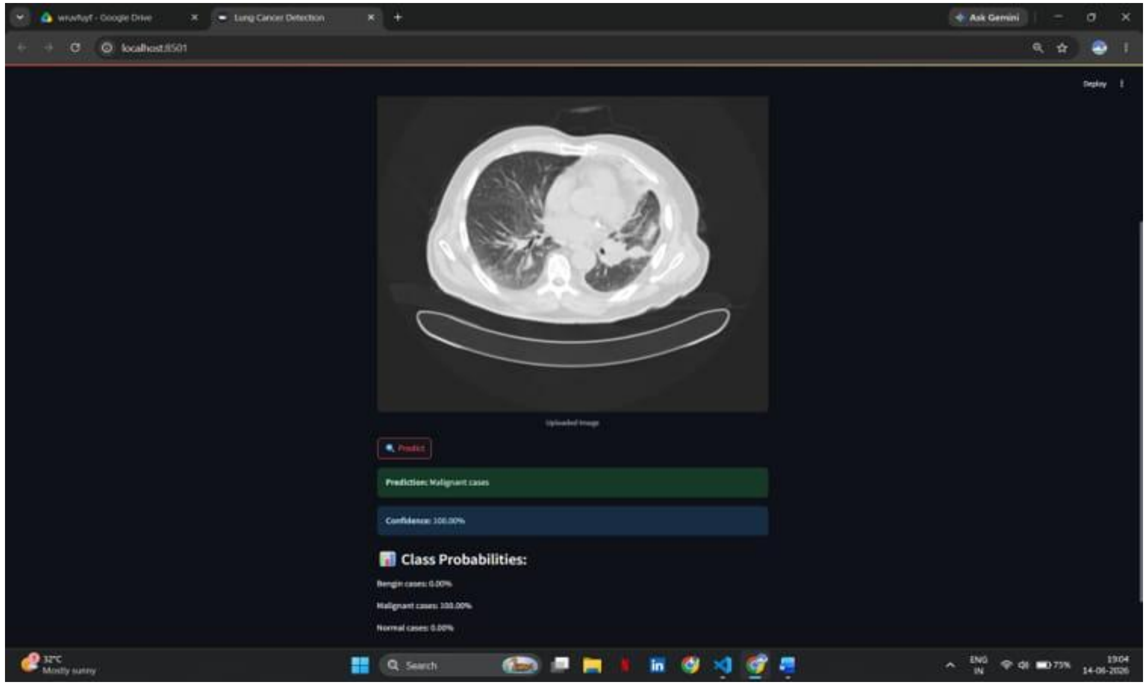
1. Accuracy graph
2. Loss graph
3. Confusion matrix
4. Sample prediction images

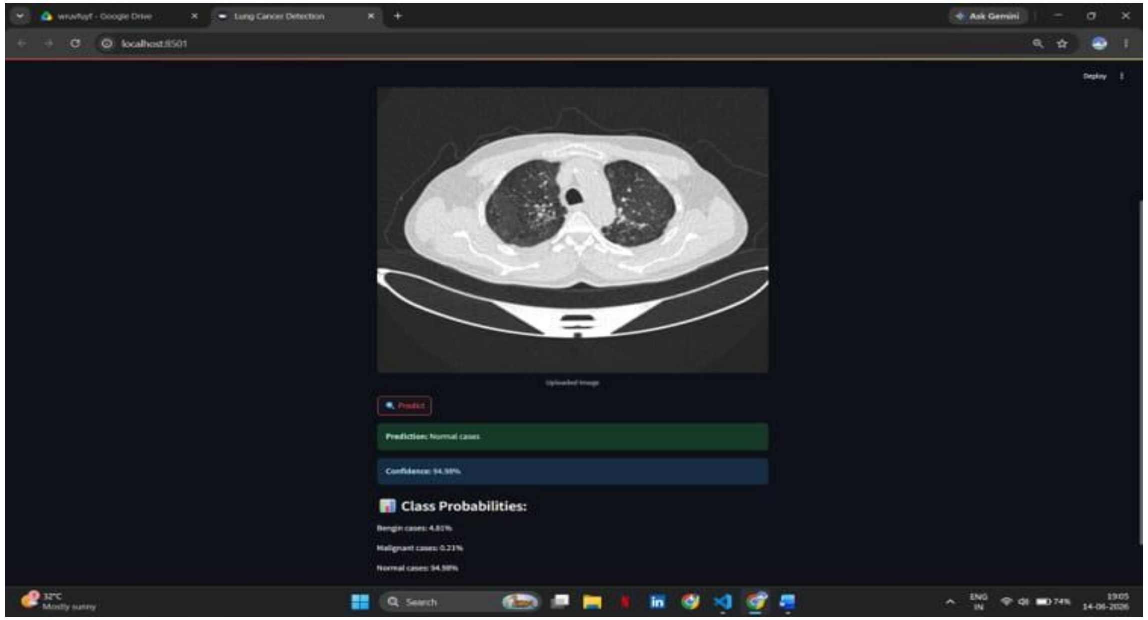

Lung Cancer Detection
Upload a lung CT scan image to predict cancer type.
Choose an image...
Drag and drop file here
Limit 200MB per file • JPG, JPEG, PNG
Browse files
Deploy


app.py
def preprocess_image(img):
img_array = img_array[:, :, :3]
img_array = np.expand_dims(img_array, axis=0)
return img_array
# MAIN LOGIC
if uploaded_file is not None:
image = Image.open(uploaded_file)
st.image(image, caption="Uploaded Image", use_container_width=True)
if st.button("Predict"):
with st.spinner("Analyzing..."):
processed_img = preprocess_image(image)
predictions = model.predict(processed_img)
predicted_class = CLASS_NAMES[np.argmax(predictions)]
confidence = np.max(predictions) * 100
# OUTPUT
st.success(f"**Prediction:** {predicted_class}")
st.info(f"**Confidence:** {confidence:.2f}%")


main.py
plt.figure()
plt.plot(history.history['accuracy'])
plt.plot(history.history['val_accuracy'])
plt.title("Accuracy")
plt.legend(["Train", "Validation"])
plt.show()
plt.figure()
plt.plot(history.history['loss'])
plt.plot(history.history['val_loss'])
plt.title("Loss")
plt.legend(["Train", "Validation"])
plt.show()
# EVALUATION
val_preds = model.predict(val_data)
val_preds_classes = np.argmax(val_preds, axis=1)
print("\nClassification Report:\n")
print(classification_report(val_data.classes, val_preds_classes))
cm = confusion_matrix(val_data.classes, val_preds_classes)
plt.figure()
sns.heatmap(cm, annot=True, fmt='d',
xticklabels=train_data.class_indices.keys(),
yticklabels=train_data.class_indices.keys())
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")


Lung Cancer Detection
localhost:8501
Deploy
Uploaded image
Predict
Prediction: Bengin cases
Confidence: 99.16%
Class Probabilities:
Bengin cases: 99.16%
Malignant cases: 0.04%
Normal cases: 0.80%


Lung Cancer Detection
localhost:8501
Deploy
Uploaded image
Predict
Prediction: Malignant cases
Confidence: 100.00%
Class Probabilities:
Bengin cases: 0.00%
Malignant cases: 100.00%
Normal cases: 0.00%


Lung Cancer Detection
localhost:8501
Deploy
Uploaded image
Predict
Prediction: Normal cases
Confidence: 94.98%
Class Probabilities:
Bengin cases: 4.81%
Malignant cases: 0.21%
Normal cases: 94.98%

CONCLUSION

- CNN and Xception provide effective lung cancer detection.
- Transfer learning improves model performance.
- AI can assist radiologists in diagnosis.
- Early detection improves patient survival.
  Demonstrates the potential of AI in healthcare.

"Artificial Intelligence is transforming medical diagnosis by enabling faster, more accurate, and intelligent healthcare solutions."